

Guhnnoo Yun

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Korea University & Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea

Spectral analysis and adaptive filtering for visual perception, from recognition to image generation.

Research Focus

My research is about visual perception for machine intelligence: I analyze what vision models pass or suppress in the spectral domain and turn that analysis into operators for visual recognition and image generation. Next, I plan to do the same for video generation and world models at a larger scale.

- Spectral analysis of convolution and self-attention (frequency balancing, graph-spectral frequency responses)
- Spectral-adaptive operators as general-purpose replacements for softmax attention, from vision backbones to diffusion transformers (SPAM, SALA)
- Visual perception for robots and healthcare (fall detection, isolation-ward services, daily-life monitoring)

Education

Ph.D., Computer Science and Engineering , Korea University	Feb 2027 (expected)
Dissertation: <i>Spectral Analysis and Adaptive Filtering for Visual Perception</i>	
M.S., Mechatronics , Gwangju Institute of Science and Technology (GIST)	2018
Thesis: <i>Removal of Jitter Noise using a Modified Kalman Filter in 3D Shape Recovery from Focus</i>	
B.S., Control and Measurement Engineering , Gyeongsang National University	2016

Research Experience

Student Researcher , Intelligence and Interaction Research Center, KIST	2019–Present
Supervisor: Dr. Dong Hwan Kim	

From spectral analysis of vision models to operators for visual recognition and image generation.

Linear Attention for Generative and Vision Transformers (SALAFormer; in preparation [P2])

- Designed **SALA** (Spectral-Adaptive Linear Attention) as closed-form associative memory: each head solves a small ridge regression for what to store in its fixed-size memory, instead of summing raw key–value products, and a learned spectral filter on the key Gram matrix allocates that memory, at linear $O(N)$ inference cost.
- **Image generation**: lower FID than MHLA (ICLR 2026), a recent linear-attention design for diffusion transformers, at both model sizes: $59.80 \rightarrow \mathbf{57.65}$ (DiT-S/2) and $37.47 \rightarrow \mathbf{36.54}$ (DiT-B/2) vs. its published results (class-conditional ImageNet 256×256 , no guidance).
- **Image classification**: SALA in a vision backbone (SALAFormer) reaches **77.2%** / **81.8%** ImageNet-1K top-1 at DeiT-Tiny/Small size, above the 76.5% / 81.6% reported for ViT³ (CVPR 2026) with the same recipe.

Spectral Analysis of Convolution and Attention for Vision Backbones

- **SPANetV2** (TPAMI 2026): compared what convolution and self-attention pass or suppress in one graph-spectral framework, showing that window size shapes their frequency responses, and turned this into a mixer with multi-scale kernels and spectral re-scaling; the backbone outperforms state-of-the-art models on ImageNet-1K, COCO, and ADE20K. [J1]
- **SPANet** (ICCV 2023): self-attention acts as a low-pass filter and convolution retains high frequencies. Prior work improved self-attention with stronger high-pass filtering; SPANet showed that stronger low-pass filtering also improves convolution, and that *balancing* high and low frequencies in a token mixer improves performance. [C3]

Applied Vision for Robotics, Healthcare & 3D Shape Recovery

- Fall detection of older adults by a healthcare robot [C5]; robot-assisted services in isolation wards [C4]; daily-life arousal monitoring and brain–computer interface decoding with egocentric vision and EEG [C1] [C2]; a view-consistent evaluation metric for multi-view panoptic segmentation [P1].
- Shape from focus, a follow-up to the M.S. thesis: 3D shape from few images with Gaussian-process regression [C6].

Intern Researcher , Center for Intelligent Robotics, KIST	2018–2019
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- Perception for healthcare robots: detecting when people with dementia leave home, to keep them from going missing [C8]; accelerometer–depth fall detection for older adults [C7], later extended with LSTMs [C5].

Publications

Published / Accepted

[J1] Guhnoo Yun; J. Yoo; K. Kim; J. Lee; P.H. Seo; D.H. Kim. “Spectral-Adaptive Modulation Networks for Visual Perception.” *IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI)*, 48(10), 11511–11525, **2026**. [arXiv] [code]

[C1] W. Hyung; M. Kim; Y.-S. Kim; J. Lee; B. Ko; Guhnoo Yun; D.H. Kim; C.-H. Im. “Context-Aware Daily-Life Arousal Monitoring Using Egocentric Vision and EEG.” *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, **2026**.

[C2] Y.-S. Kim; M. Kim; W. Hyung; B. Ko; Guhnoo Yun; D.H. Kim; C.-H. Im. “PeBCI: Vision-Guided Brain-Computer Interface for Context-Aware Neural Decoding in Daily Life.” *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, **2026**.

[C3] **Guhnnoo Yun**; J. Yoo; K. Kim; J. Lee; D.H. Kim. "SPANet: Frequency-balancing Token Mixer using Spectral Pooling Aggregation Modulation." *IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 6090–6101, **2023**. [CVF] [arXiv] [project] [code]

[C4] Y. Kwon; S. Shin; K. Yang; S. Park; S. Shin; H. Jeon; K. Kim; **Guhnnoo Yun**; ...; Y. Lim. "Heterogeneous Robot-Assisted Services in Isolation Wards: A System Development and Usability Study." *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 8069–8076, **2023**.

[C5] K. Kim; **Guhnnoo Yun**; S.-K. Park; D.H. Kim. "Efficient Fall Detection for a Healthcare Robot System Based on 3-Axis Accelerometer and Depth Sensor Fusion with LSTM Networks." *Int. Conf. on Pattern Recognition (ICPR)*, 2207–2212, **2022**.

[C6] H.-S. Jang; **Guhnnoo Yun**; M.T. Mahmood; M.-K. Kang. "Optimal Sampling for Shape from Focus by Using Gaussian Process Regression." *IEEE Int. Conf. on Consumer Electronics (ICCE)*, 1–4, **2020**.

[C7] K. Kim; **Guhnnoo Yun**; S.-K. Park; D.H. Kim. "Fall Detection for the Elderly Based on 3-Axis Accelerometer and Depth Sensor Fusion with Random Forest Classifier." *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 4611–4614, **2019**.

[C8] **Guhnnoo Yun**; K. Kim; S.-K. Park; D.H. Kim. "A Monitoring System to Support Home Health Care for the Elderly with Dementia by Detecting Going Out Activities Based on RGB-D Sensors." *Int. Conf. on Ubiquitous Robots (UR)*, 71–76, **2019**.

Under Review

[P1] Y. Lee; B. Ko; **Guhnnoo Yun**; D.H. Kim. "Every View Counts: View-Consistent Panoptic Quality for Multi-view Panoptic Segmentation." *arXiv preprint arXiv:2610.05911*, **2026**.

In Preparation

[P2] **Guhnnoo Yun** et al. "SALAFormer: Spectral-Adaptive Linear Attention as Closed-Form Associative Memory."

Awards & Scholarships

Outstanding Student Researcher Scholarship, KIST	2026
Outstanding Technology Award, KIST	2023
Presidential Award, 2020 International Robot Contest—Humanoid Robot Sports (Deep Learning Algorithm), Ministry of the Interior and Safety	2020

Invited Talks

"Recent Research Trends in Computer Vision," seminar at Korea Polytechnic University (now Tech University of Korea)	2021
"6D Object Pose Estimation," seminar series at CUBOX (now SECERN AI)	2021

Professional Service

Reviewer, International Conference on Pattern Recognition (ICPR)	2024, 2026
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Technical Skills

Languages & Frameworks: Python, PyTorch, C++, MATLAB; timm, MMDetection, MMSegmentation, ROS

References

Dr. Dong Hwan Kim KIST gregorykim@kist.re.kr	Prof. Soohwan Kim Kwangwoon University kimsoohwan@kw.ac.kr	Dr. Yoonseob Lim KIST yslim@kist.re.kr
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